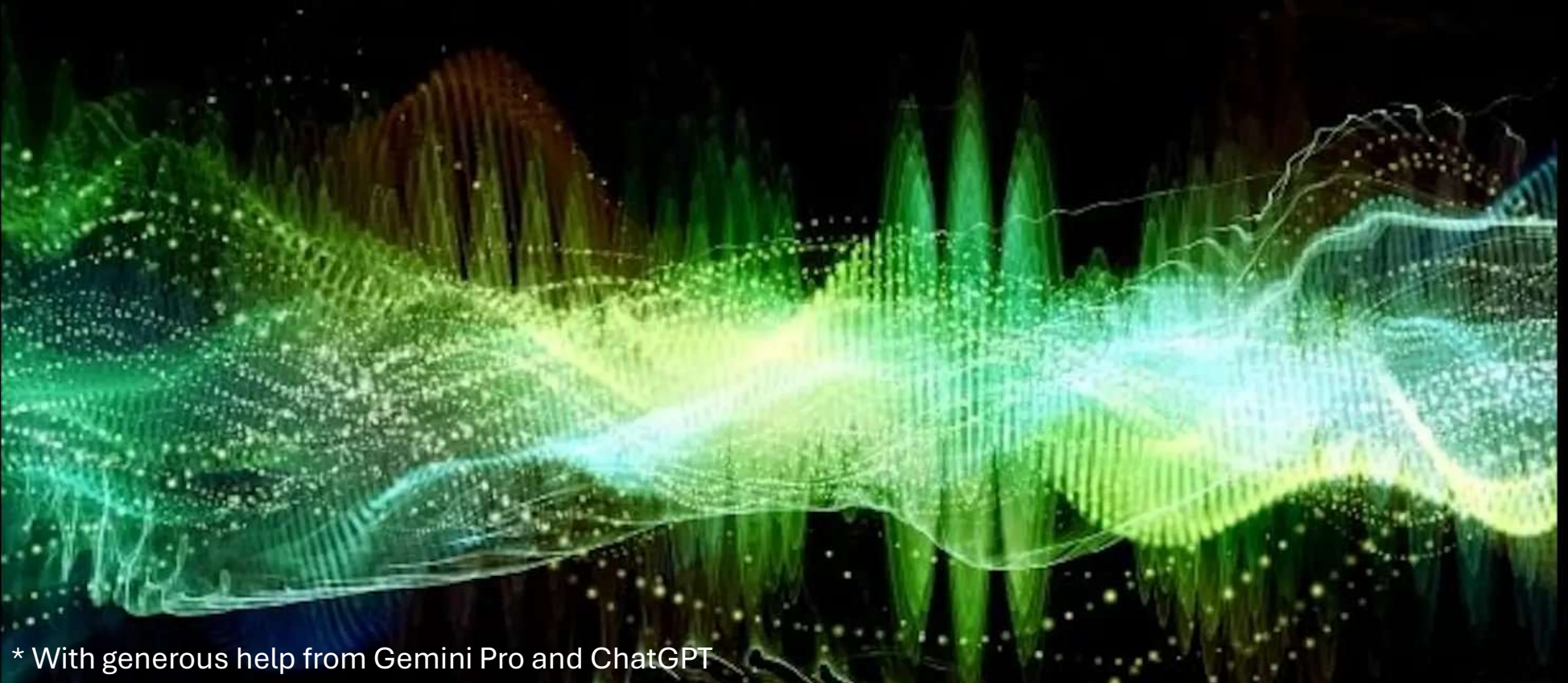


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 9, 9-22-26, 9-29-26

Peter Mathys*



* With generous help from Gemini Pro and ChatGPT

Time/Frequency Responses and Group Delay

From pole-zero geometry to waveform dispersion and all-pass phase equalization

pole-zero geometry → group delay → all-pass → ML feature timing

75 minutes • senior + first-year graduate ECE

Equations typeset in standard mathematical notation

Roadmap

- 0–15 min • Read magnitude and phase directly from pole-zero geometry.
- 15–40 min • Distinguish phase delay from group delay and connect nonlinear phase to dispersion.
- 40–60 min • Build all-pass systems from reciprocal pole-zero structure.
- 60–75 min • Ask whether preprocessing preserves the timing features an ML model actually uses.

Frequency response is the z-transform evaluated on the unit circle

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$$

- Every digital frequency ω corresponds to one point $e^{j\omega}$ on the unit circle.
- Magnitude and phase can be read geometrically from vectors drawn from poles and zeros to that point.

Pole-zero geometry: magnitude is a distance ratio

$$|H(e^{j\omega})| = |K| \frac{\prod_i |e^{j\omega} - z_i|}{\prod_k |e^{j\omega} - p_k|}$$

Interpretation

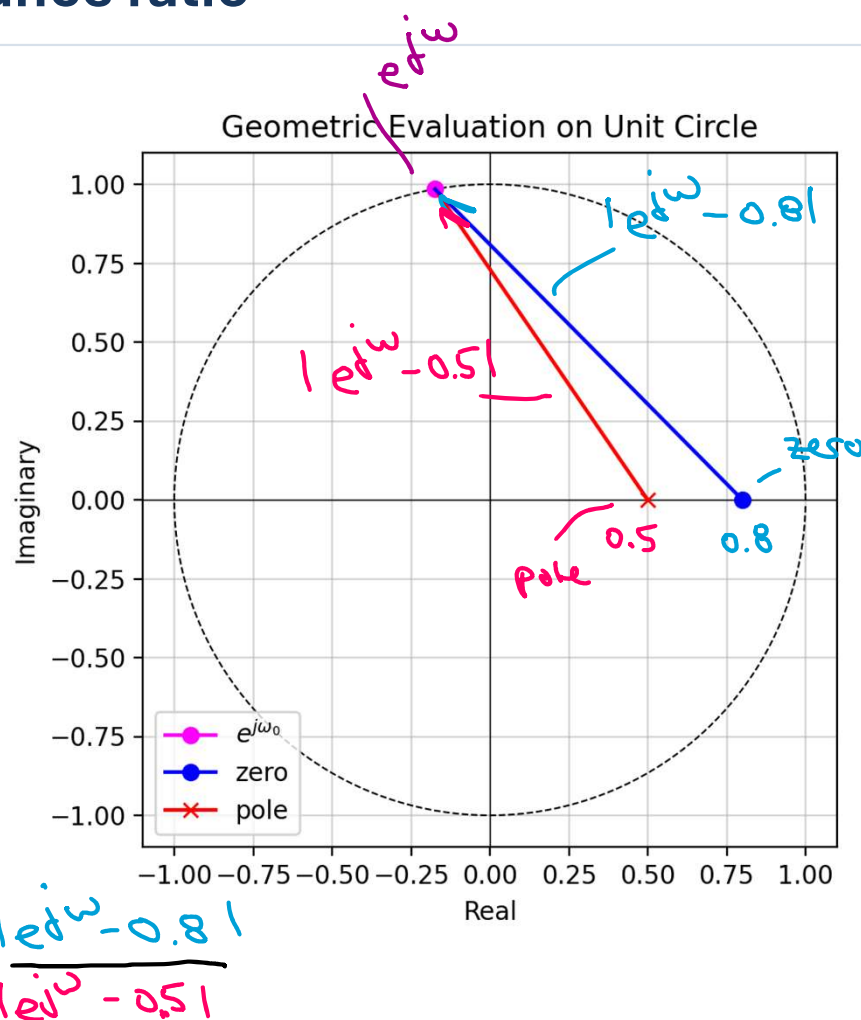
Zeros pull the response toward zero when the unit-circle point passes nearby. Poles push the response upward when the unit-circle point passes nearby.

$$H(z) = \frac{1 - 0.8z^{-1}}{1 - 0.5z^{-1}}$$

$$\text{zero } z = 0.8$$

$$\text{pole } z = 0.5$$

$$|H(e^{j\omega})| = \left| \frac{1 - 0.8e^{-j\omega}}{1 - 0.5e^{-j\omega}} \right| = \frac{|e^{j\omega} - 0.8|}{|e^{j\omega} - 0.5|} = \frac{|e^{j\omega} - 0.8|}{|e^{j\omega} - 0.5|}$$



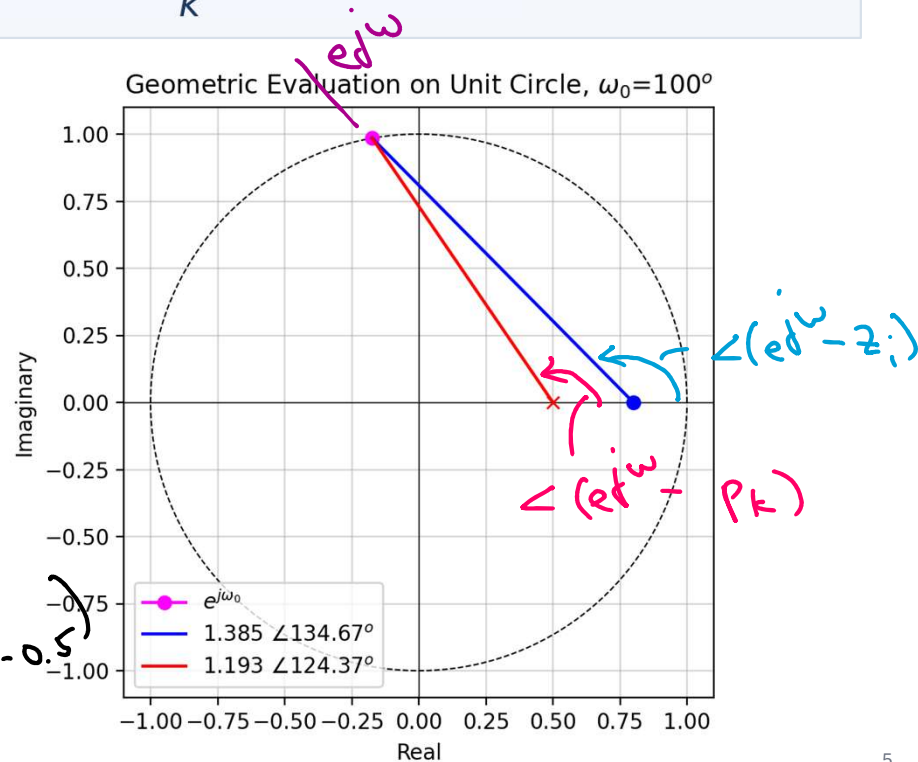
Pole-zero geometry: phase is an angle sum

$$\angle H(e^{j\omega}) = \angle K + \sum_i \angle(e^{j\omega} - z_i) - \sum_k \angle(e^{j\omega} - p_k)$$

- Each zero contributes a vector angle.
- Each pole contributes a vector angle with the opposite sign.
- Phase can vary rapidly even when magnitude looks smooth.

$$H(e^{j\omega}) = \frac{e^{j\omega} - 0.8}{e^{j\omega} - 0.5}$$

$$\angle H(e^{j\omega}) = \tan^{-1}\left(\frac{\sin \omega}{\cos \omega - 0.8}\right) - \tan^{-1}\left(\frac{\sin \omega}{\cos \omega - 0.5}\right)$$



Worked geometry example

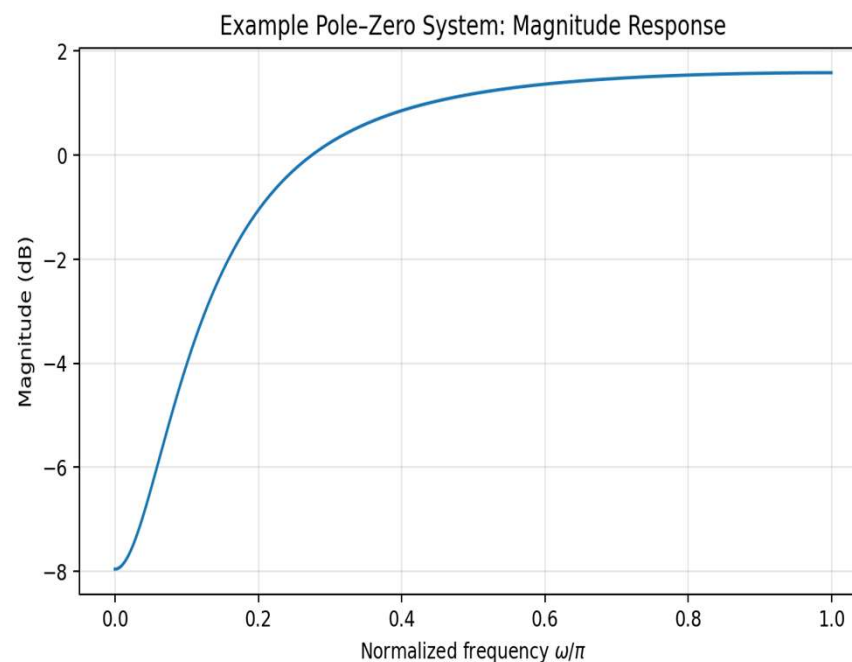
$$H(z) = \frac{1 - 0.8z^{-1}}{1 - 0.5z^{-1}}$$

At $\omega=0$

The unit-circle point is $z=1$. The zero-distance is 0.2 and pole-distance is 0.5, so $|H(1)|=0.4$.

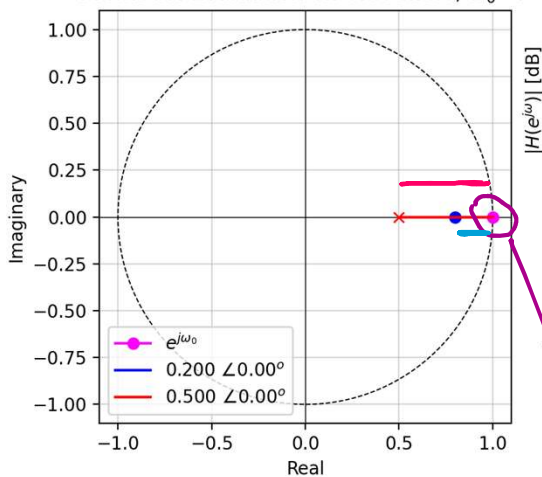
At $\omega=\pi$

The point is $z=-1$. The distances are 1.8 and 1.5, so $|H(-1)|=1.2$. Predict a high-frequency tilt before plotting.



Worked geometry example

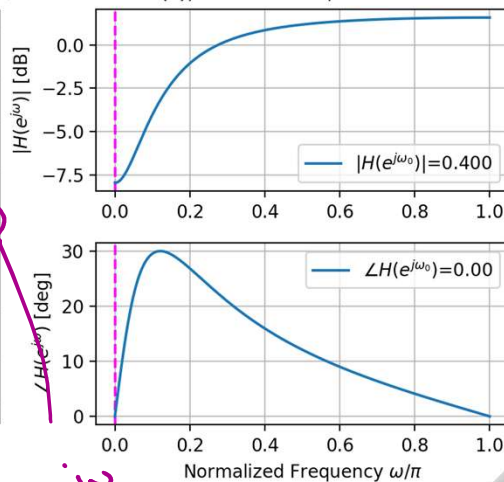
Geometric Evaluation on Unit Circle, $\omega_0=0^\circ$



$$\omega = 0$$

Small blue
large red

$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$



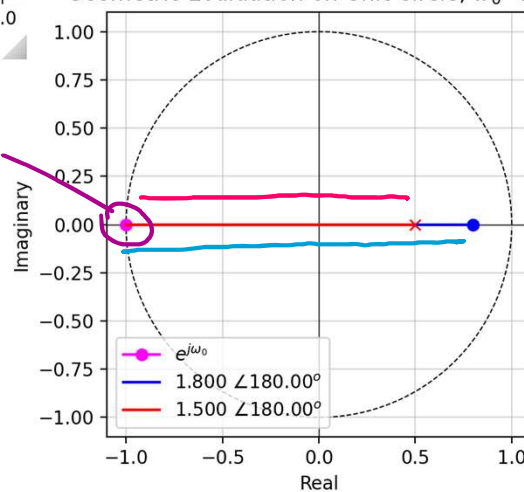
$\pi\omega$

$\pi\omega$

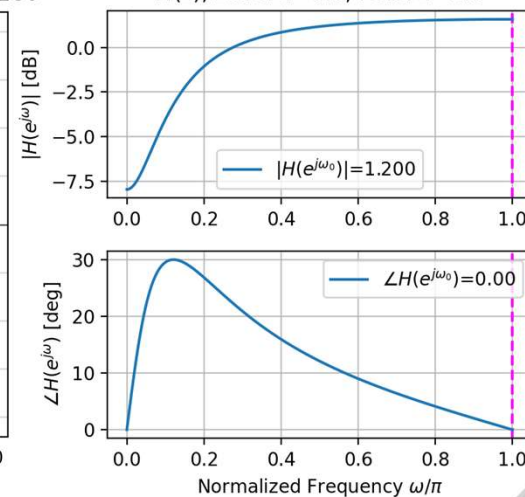
large blue
smaller red

$$\omega = \pi$$

Geometric Evaluation on Unit Circle, $\omega_0=180^\circ$

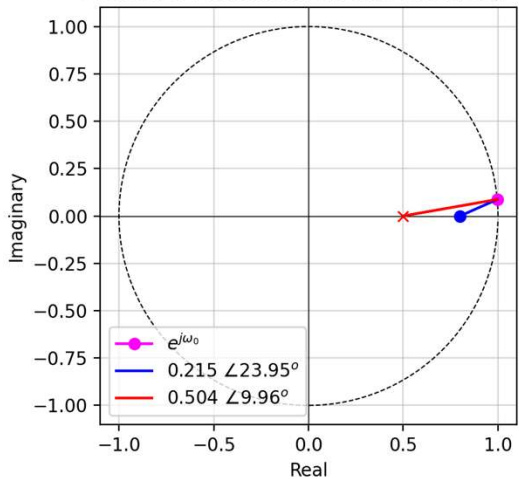


$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$

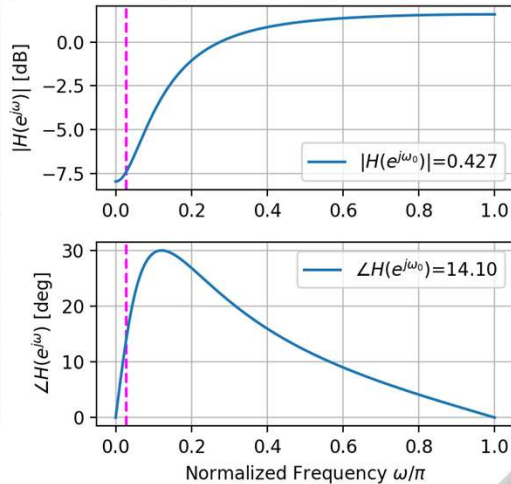


Worked geometry example

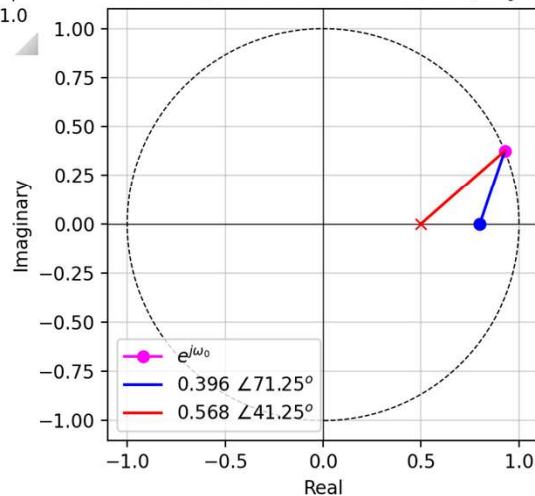
Geometric Evaluation on Unit Circle, $\omega_0=5^\circ$



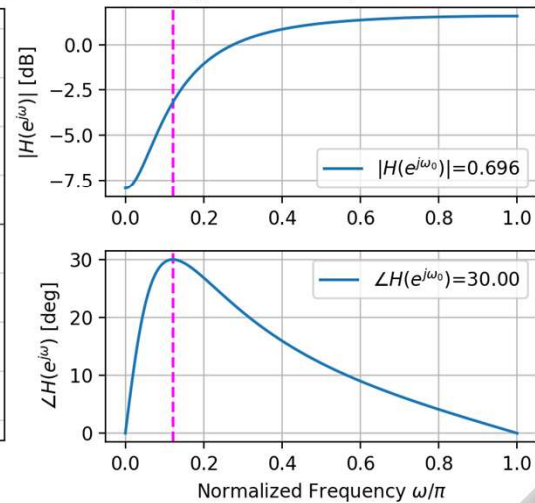
$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$



Geometric Evaluation on Unit Circle, $\omega_0=22^\circ$

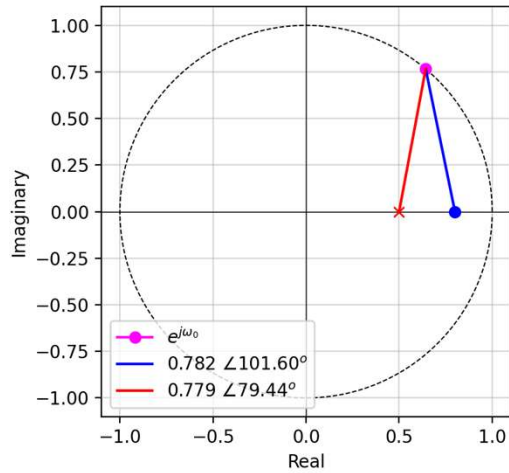


$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$

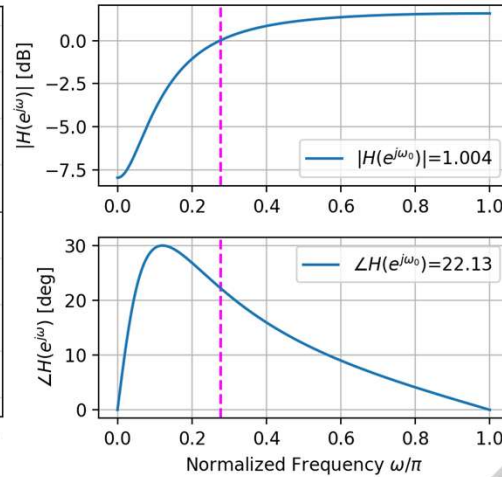


Worked geometry example

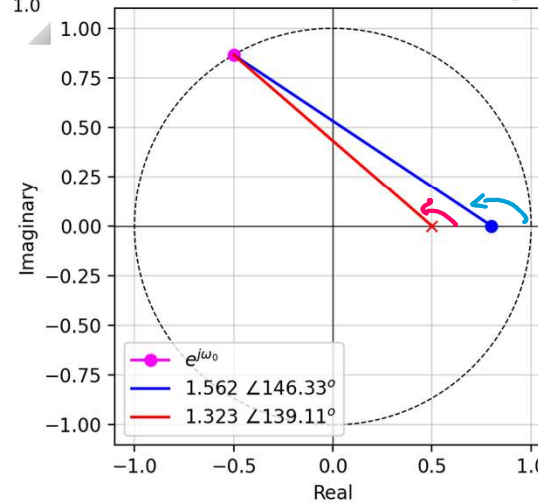
Geometric Evaluation on Unit Circle, $\omega_0=50^\circ$



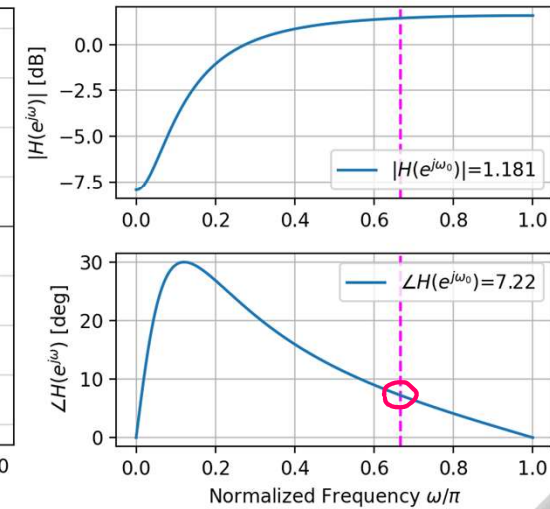
$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$



Geometric Evaluation on Unit Circle, $\omega_0=120^\circ$



$H(z)$, Zero: $z=0.8$, Pole: $z=0.5$



Concept check: move a pole toward the unit circle

- A pole at angle ω_0 moves radially from radius 0.5 to radius 0.95. What happens near ω_0 ?

Concept check: move a pole toward the unit circle

- A pole at angle ω_0 moves radially from radius 0.5 to radius 0.95. What happens near ω_0 ?
- Good answer: the denominator vector becomes much shorter, so magnitude increases sharply and the phase changes more rapidly.
- Graduate-level addition: the pole also produces a more concentrated group-delay contribution near its angle.

Phase delay and group delay answer different questions

$$\tau_p(\omega) = -\frac{\angle H(e^{j\omega})}{\omega}$$

$$\tau_g(\omega) = -\frac{d}{d\omega} \angle H(e^{j\omega})$$

Phase delay

For a single sinusoidal component, how much phase corresponds to an equivalent delay?

Group delay

For a narrowband packet, how does the envelope / modulation arrive versus frequency?

Do not conflate them

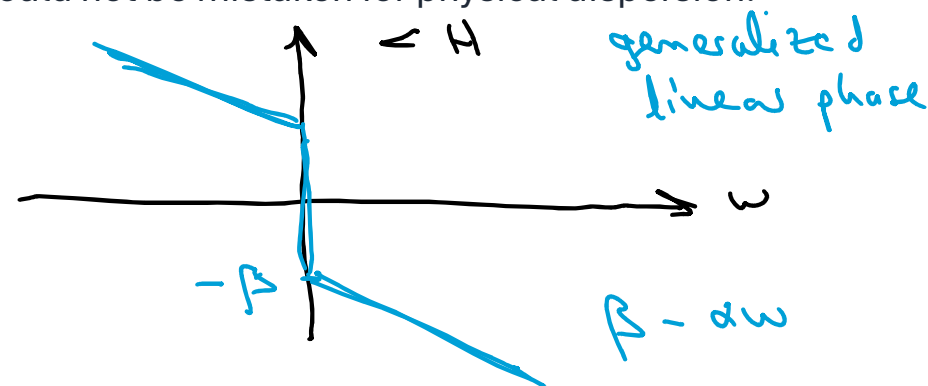
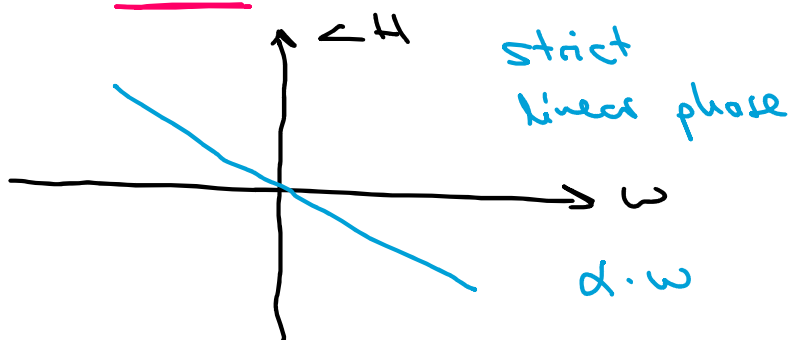
They are equal for pure linear phase through the origin, but generally differ.

Linear phase gives constant group delay

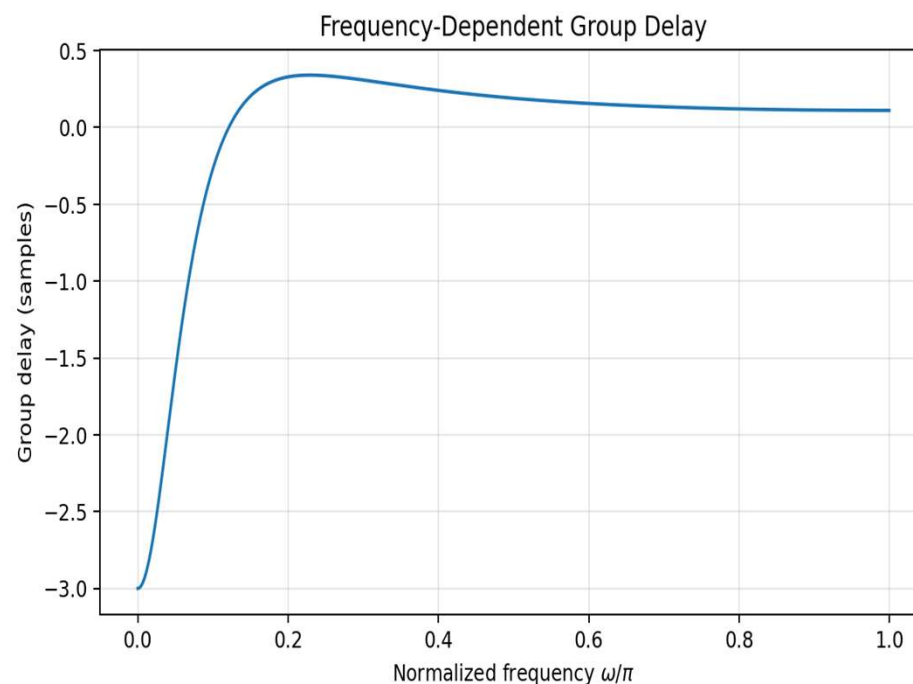
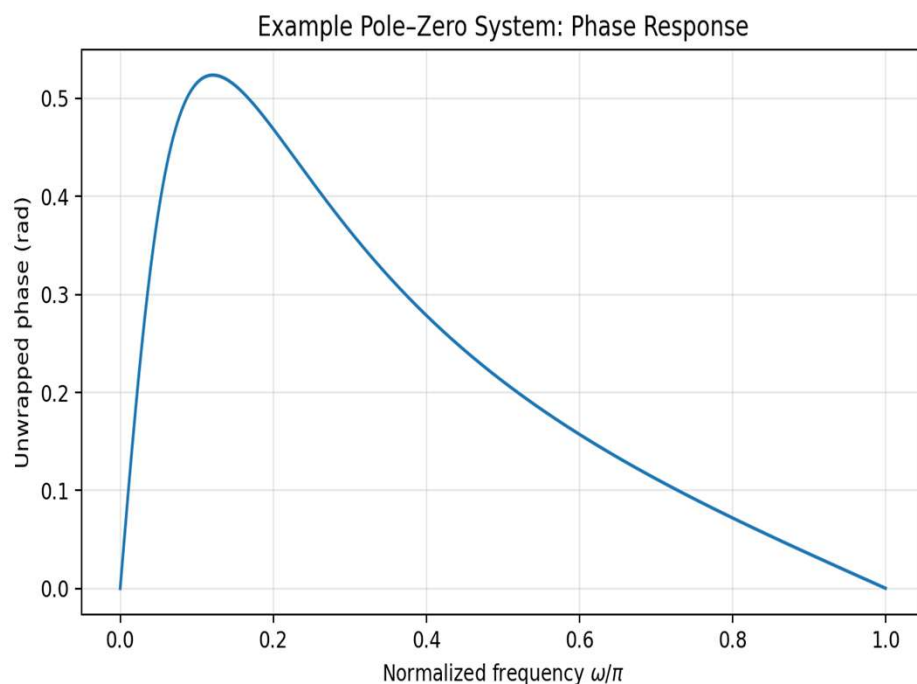
$$\angle H(e^{j\omega}) = \beta - \alpha\omega$$

$$\tau_g(\omega) = \alpha$$

- All passband frequency components receive the same local delay.
- The constant β can shift phase without changing the group-delay slope.
- Possible π jumps from a real amplitude sign change should not be mistaken for physical dispersion.



Nonlinear phase produces frequency-dependent delay



Read both plots

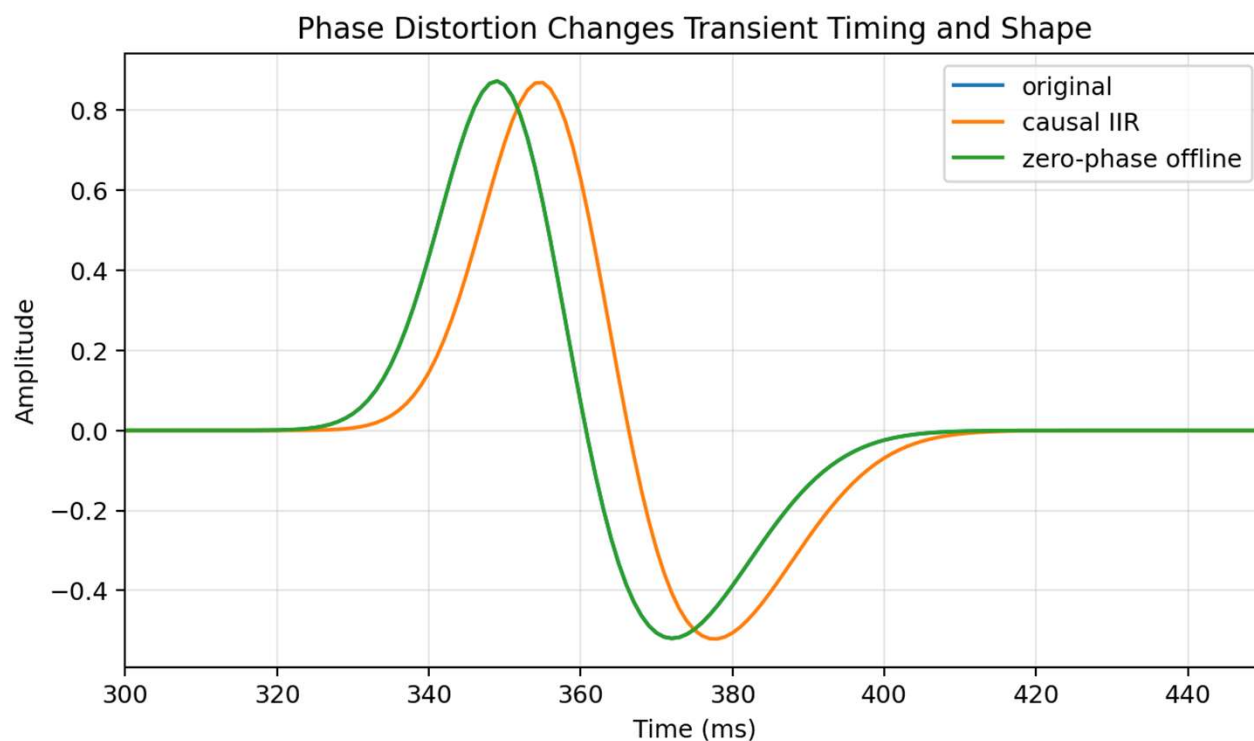
The phase curvature becomes group-delay variation. The derivative makes rapid phase changes easier to interpret as timing dispersion.

A transient is a superposition of many frequencies

- A sharp onset or pulse requires many spectral components.
- If each component experiences a different delay, the components no longer reassemble at the same relative times.
- The result can be peak shift, ringing, smearing, or altered onset slope—even if $|H(e^{j\omega})|$ looks acceptable.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

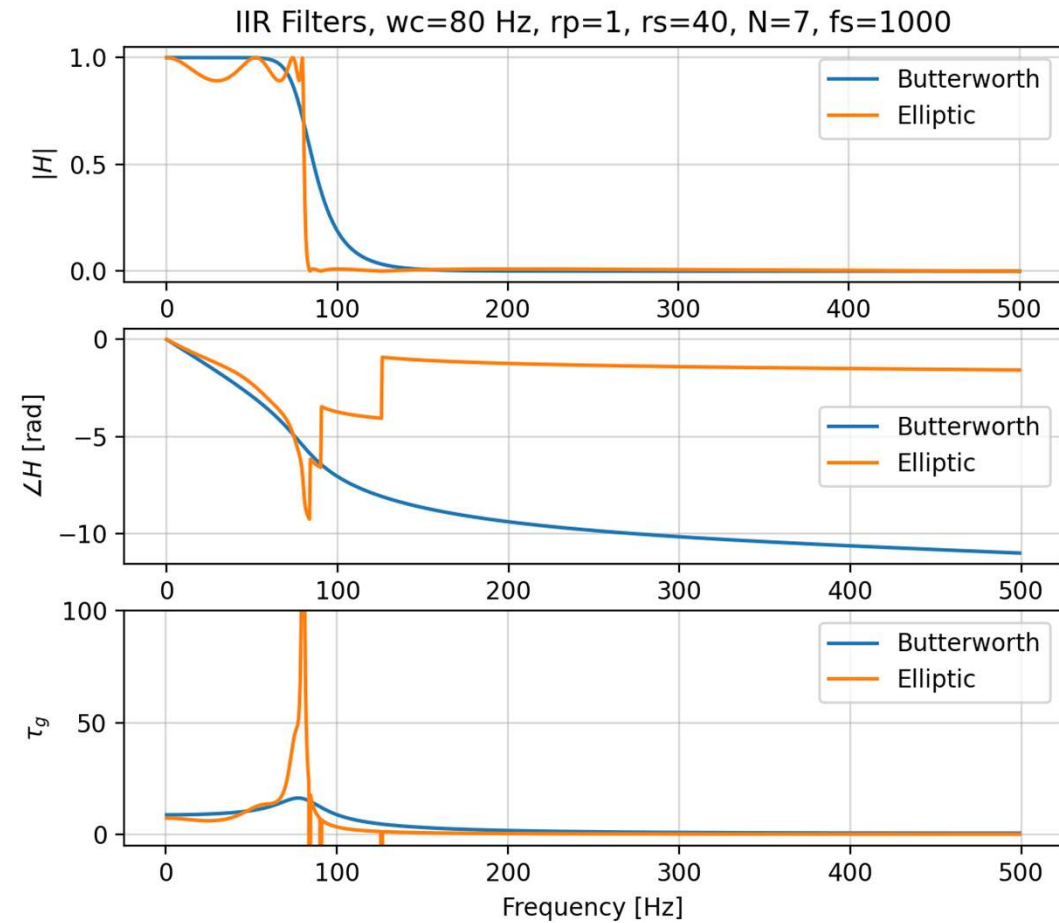
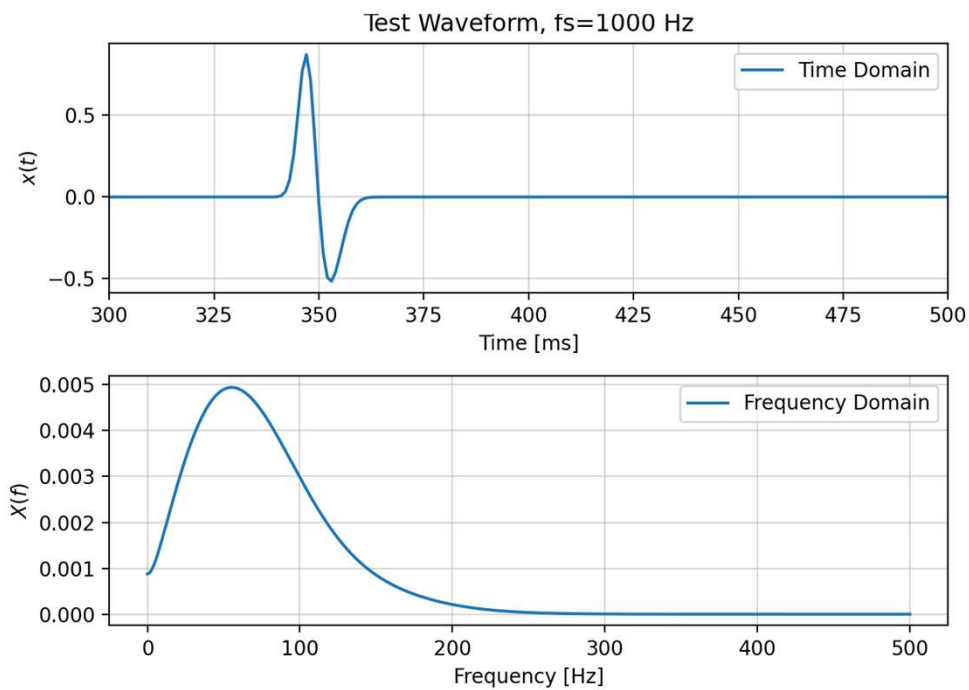
Demonstration: causal IIR dispersion versus zero-phase offline filtering



Good observation

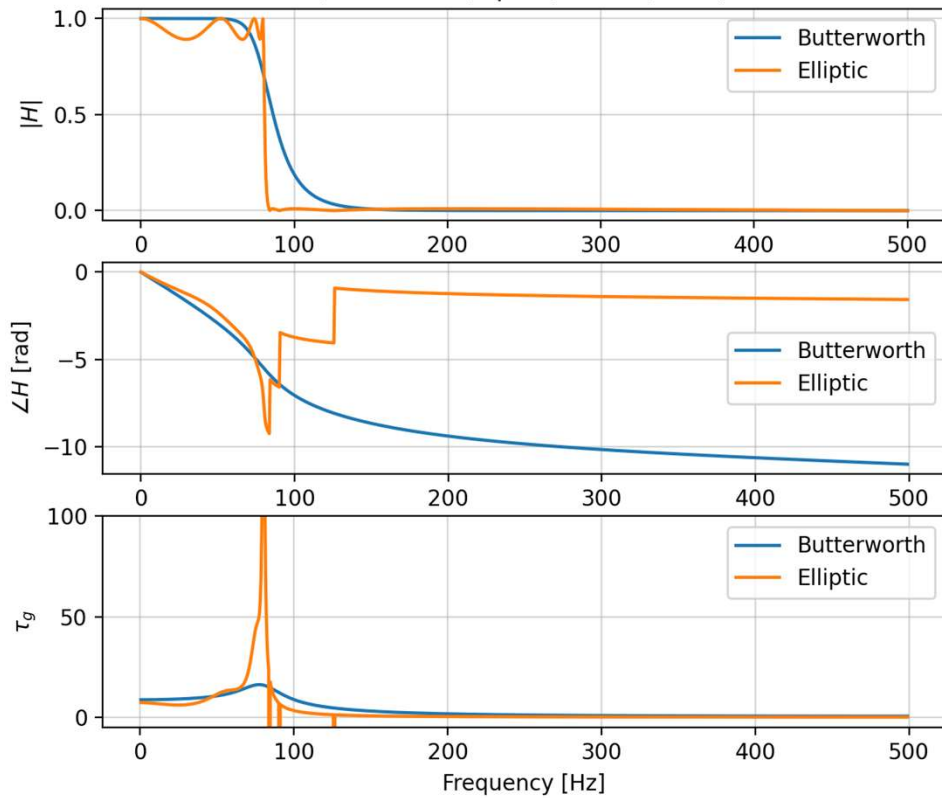
The causal IIR can shift and reshape the transient. Forward-backward filtering restores timing much better, but it is noncausal and changes the effective magnitude response.

Causal IIR dispersion versus zero-phase offline filtering

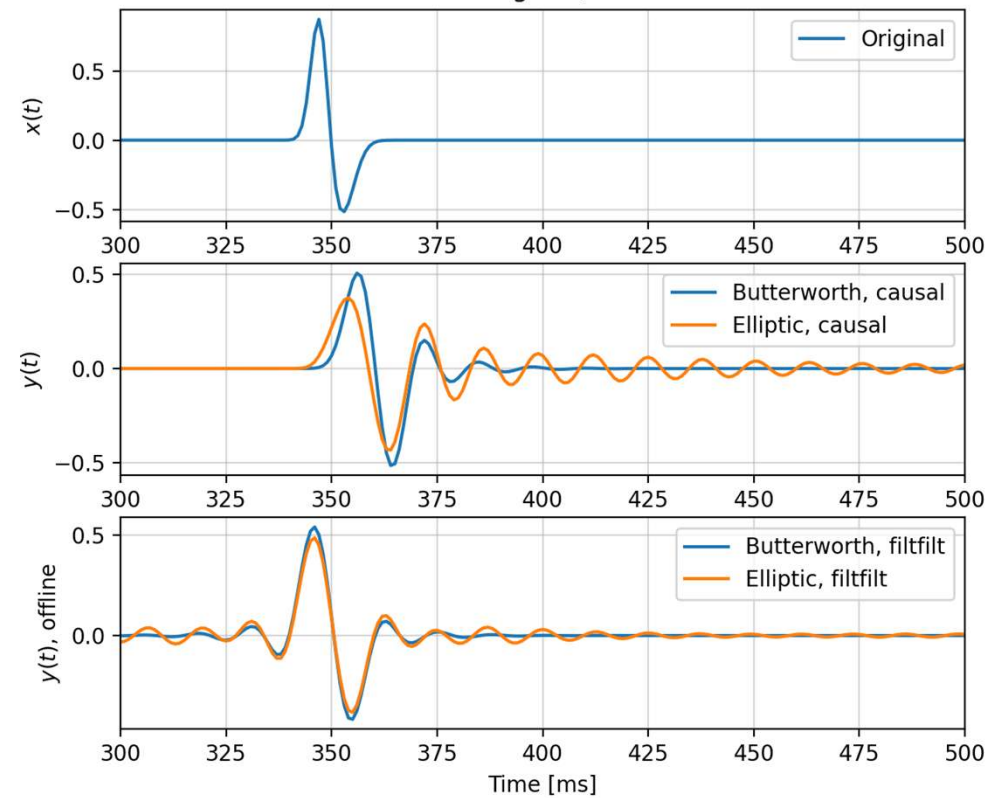


Causal IIR dispersion versus zero-phase offline filtering

IIR Filters, $w_c=80$ Hz, $r_p=1$, $r_s=40$, $N=7$, $f_s=1000$



Filtered Testsignals, $f_s=1000$ Hz



All-pass systems: magnitude one does not mean “does nothing”

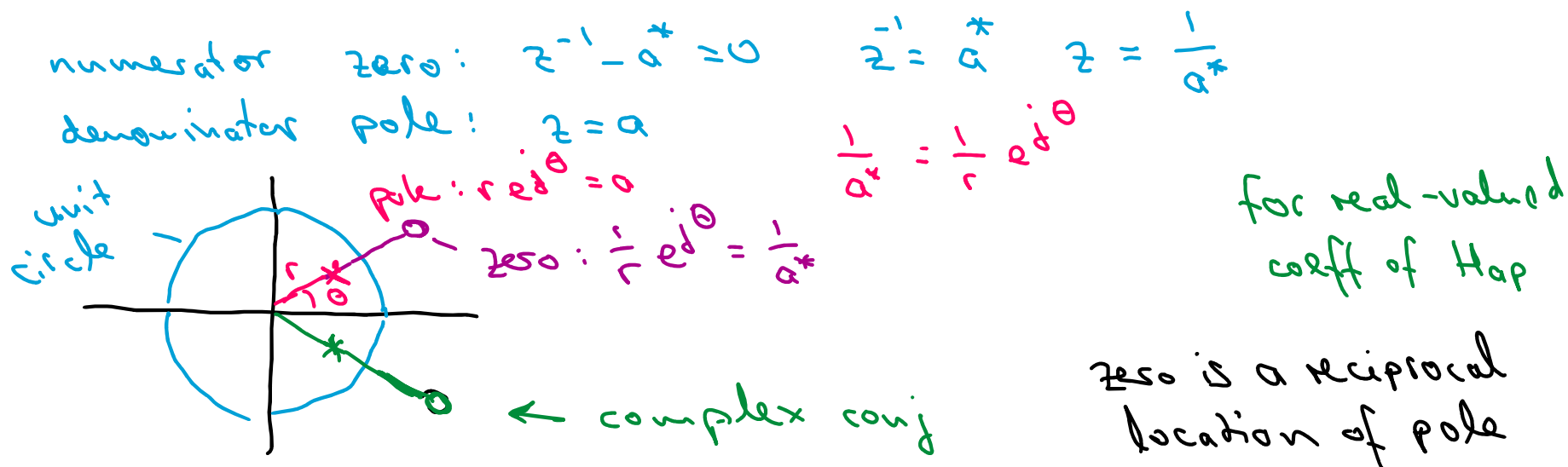
$$|H_{ap}(e^{j\omega})| = 1 \quad \forall \omega$$

- An all-pass changes phase without changing magnitude.
- That makes it a phase / group-delay design element rather than a magnitude-shaping element.
- Useful for phase equalization, fractional-delay structures, and decomposing a system into minimum-phase and all-pass parts.

First-order all-pass structure

$$H_{ap}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}, \quad |a| < 1$$

- Pole at $z=a$ lies inside the unit circle.
- Zero at $z=1/a^*$ lies at the conjugate reciprocal location outside the unit circle.
- On $|z|=1$, numerator and denominator distances match, so magnitude is unity.



Phase equalization with an all-pass

- Suppose a magnitude response is acceptable but group delay is uneven.
- Cascading an all-pass leaves total magnitude unchanged because $|H_{\text{tot}}| = |H||H_{\text{ap}}| = |H|$.
- Choose the all-pass phase so its group-delay contribution makes the total delay flatter over the band of interest.

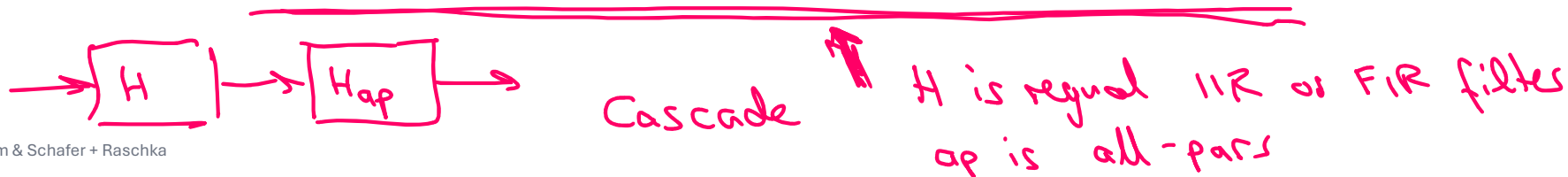
Magnitude of 1: $H_{\text{ap}}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$

$H_{\text{ap}}(e^{j\omega}) = \frac{e^{-j\omega} - a^*}{1 - ae^{-j\omega}} = e^{-j\omega} \frac{1 - a^* e^{j\omega}}{1 - ae^{-j\omega}}$

take magnitude $|H_{\text{ap}}(e^{j\omega})| = |e^{-j\omega}| \frac{|1 - a^* e^{j\omega}|}{|1 - ae^{-j\omega}|} = 1$

$\angle H_{\text{ap}}(e^{j\omega}) = \tan^{-1} \frac{\text{Im}[num]}{\text{Re}[num]} - \tan^{-1} \frac{\text{Im}[den]}{\text{Re}[den]}$ highly nonlinear

$$\tau_{g, \text{tot}}(\omega) = \tau_{g, H}(\omega) + \tau_{g, \text{ap}}(\omega)$$



What can an all-pass equalizer not fix?

- It cannot repair magnitude notches, stopband leakage, or insufficient bandwidth because its magnitude is exactly one.
- It can only redistribute phase / group delay.
- A causal stable all-pass generally adds nonnegative group delay; “flattening” often means adding delay where the original system was too fast rather than subtracting delay where it was too slow.

ML context: feature distortion happens before the model

- ECG anomaly detection: QRS peak timing and slope may be predictive.
- Audio event detection: onset rise time and inter-channel timing may be predictive.
- Vibration fault detection: impulse spacing and envelope timing may be predictive.
- A preprocessing filter that disperses these structures changes the features before an SVM, RF, MLP, or CNN can use them.

Choose your preprocessing filters so that relevant features for ML algorithm are preserved

Good ML engineering question

Ask before filtering

Does the downstream task depend primarily on long-term spectral magnitude, or on timing / waveform morphology?

- If magnitude statistics dominate, nonlinear phase may be harmless.
- If local timing is predictive, preserve phase or make training/deployment use the same causal distortion.
- Do not assume “denoised” means “more informative for the classifier.”

Python demo: inspect response and delay

```
b = np.array([1.0, -0.8])
a = np.array([1.0, -0.5])

w, H = signal.freqz(b, a, worN=2048)
phase = np.unwrap(np.angle(H))
w_gd, gd = signal.group_delay((b, a), w=2048)

# Pole-zero geometry
zeros = np.roots(b)
poles = np.roots(a)
```

```
# Compare causal IIR and zero-phase offline filtering
```

```
y_causal = signal.sosfilt(sos, x)
```

```
y_zero = signal.sosfiltfilt(sos, x)
```

Handwritten notes:

- ← IIR real-time filtering* (points to `signal.sosfilt(sos, x)`)
- ← filtfilt zero phase filtering (off-line)* (points to `signal.sosfiltfilt(sos, x)`)
- sos* (points to `sos` in both function calls)
- second order systems*
- for numerical stability, esp IIR filters*

Concept check

- Two filters have identical $|H(e^{j\omega})|$ but different $\angle H(e^{j\omega})$. Must their outputs be identical for every input?

Concept check

- Two filters have identical $|H(e^{j\omega})|$ but different $\angle H(e^{j\omega})$. Must their outputs be identical for every input?
- Good answer: no. Their spectra have the same magnitudes but different phases, so transient waveform shapes can differ.
- When would they sound or look nearly equivalent? When the task or perception is insensitive to the altered phase relationships.

Lecture 9 synthesis

- Pole-zero distance ratios explain magnitude; vector-angle sums explain phase.
- Group delay turns phase slope into a timing diagnostic.
- All-pass systems have unit magnitude but can strongly reshape phase and delay.
- ML preprocessing must preserve—or consistently reproduce—the temporal features the model needs.